Illustration

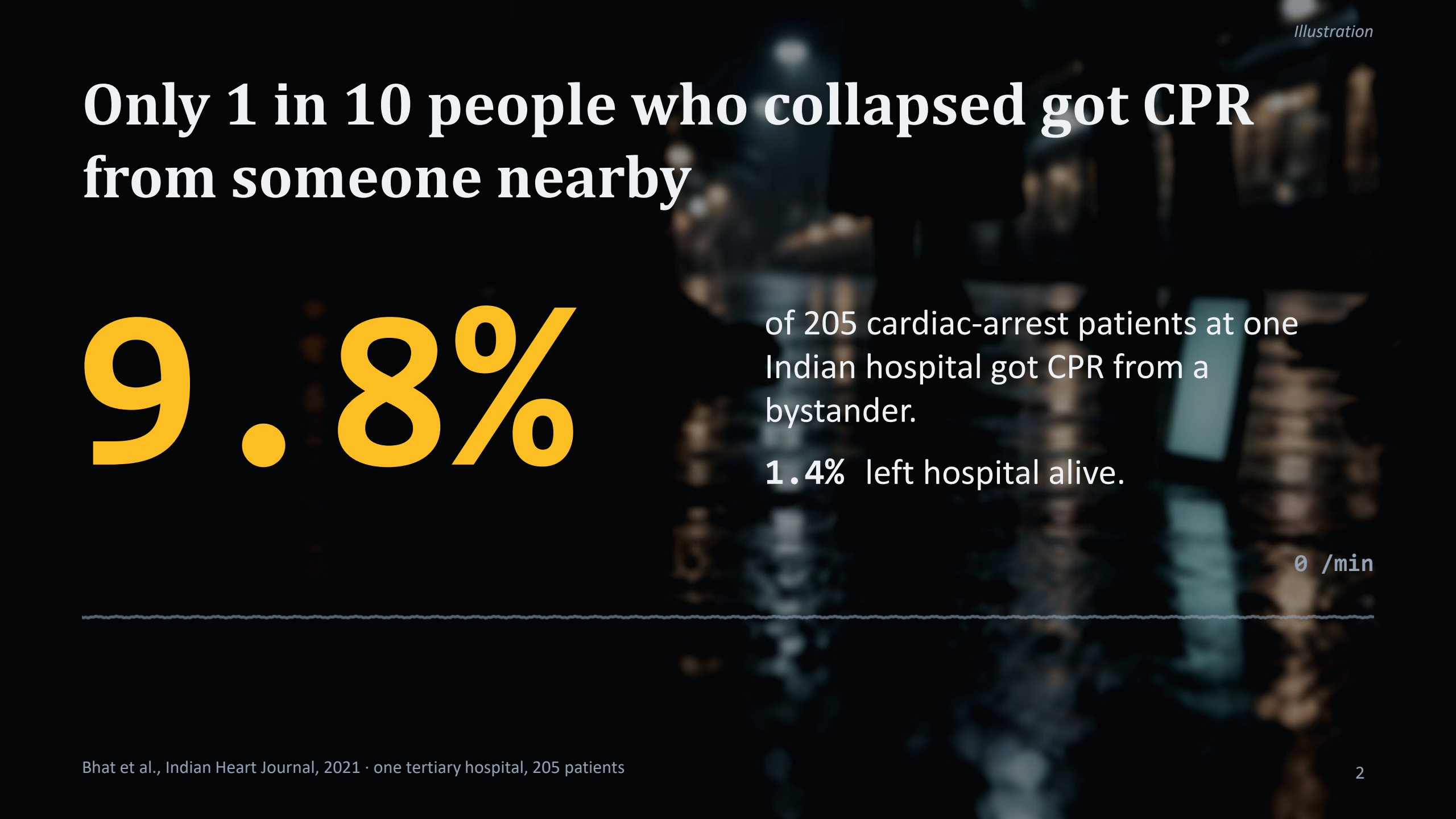
CPR Coach

A phone that watches your CPR and tells you what to fix.



Team Zero Flux Vivek · Pratham · Prabal · Moksh

ANVATHON 2026 · OPEN INNOVATION



Illustration

Only 1 in 10 people who collapsed got CPR from someone nearby

9.8%

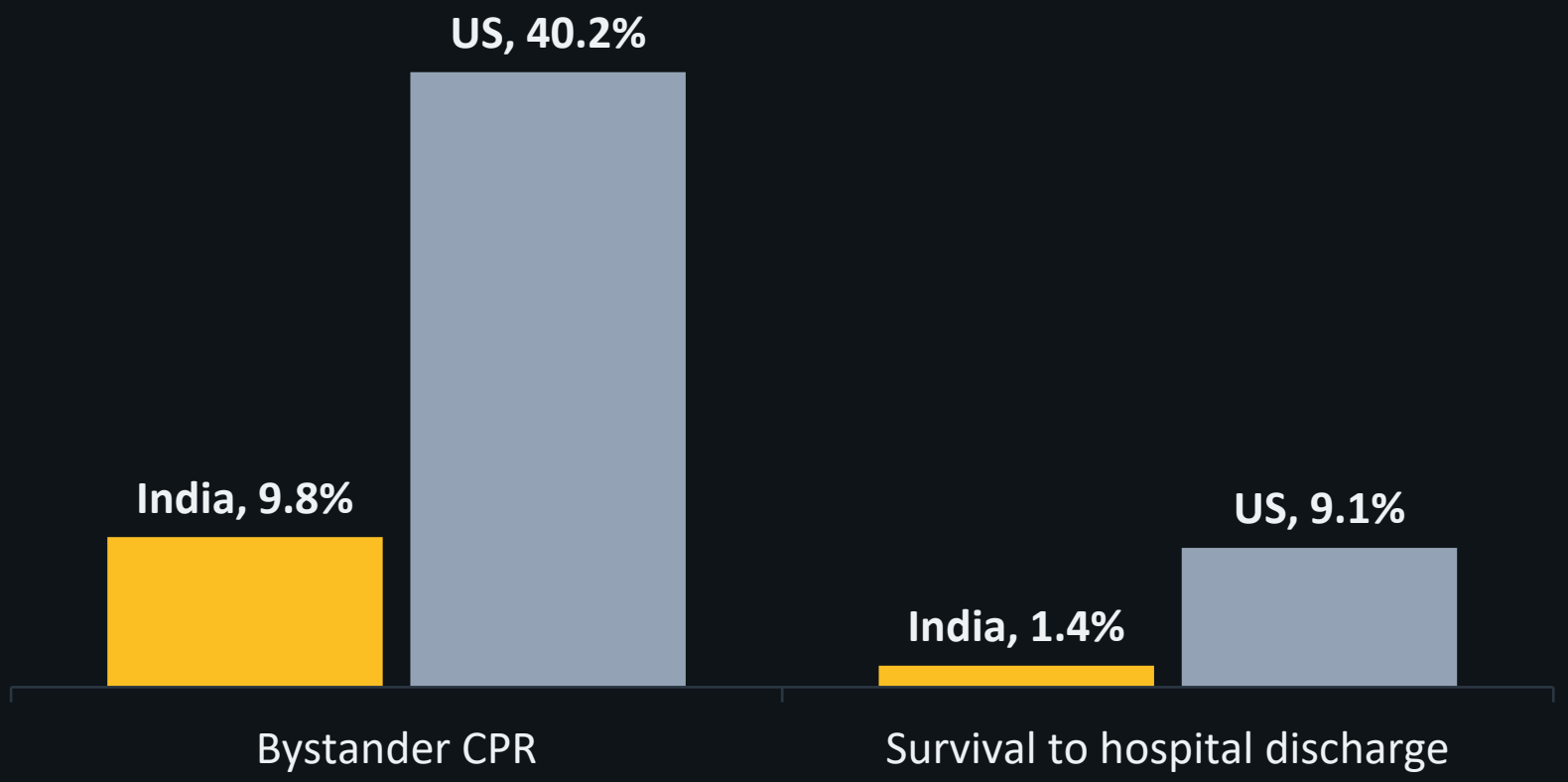
of 205 cardiac-arrest patients at one Indian hospital got CPR from a bystander.

1.4% left hospital alive.

`0 /min`

Bhat et al., Indian Heart Journal, 2021 · one tertiary hospital, 205 patients

India's bystander CPR rate is a quarter of the US rate



India
one hospital, 205 patients

US
national registry

Fewer people nearby start CPR, and fewer patients leave hospital alive.

India: Bhat et al., Indian Heart Journal 2021, one tertiary hospital, 205 patients. The two sources are not measured the same way: one hospital vs a national registry.

US: American Heart Association, Heart Disease and Stroke Statistics 2023. Bystander CPR: over 356,000 out-of-hospital arrests. Survival: 2021 US data, EMS-treated adult non-traumatic arrests.

Under 2 in 100 people say they know CPR, and the skill fades in months

<2%

of about 1 lakh people surveyed in 20 Indian cities said they knew CPR

3 months

is how soon CPR skills can start to fade after a course

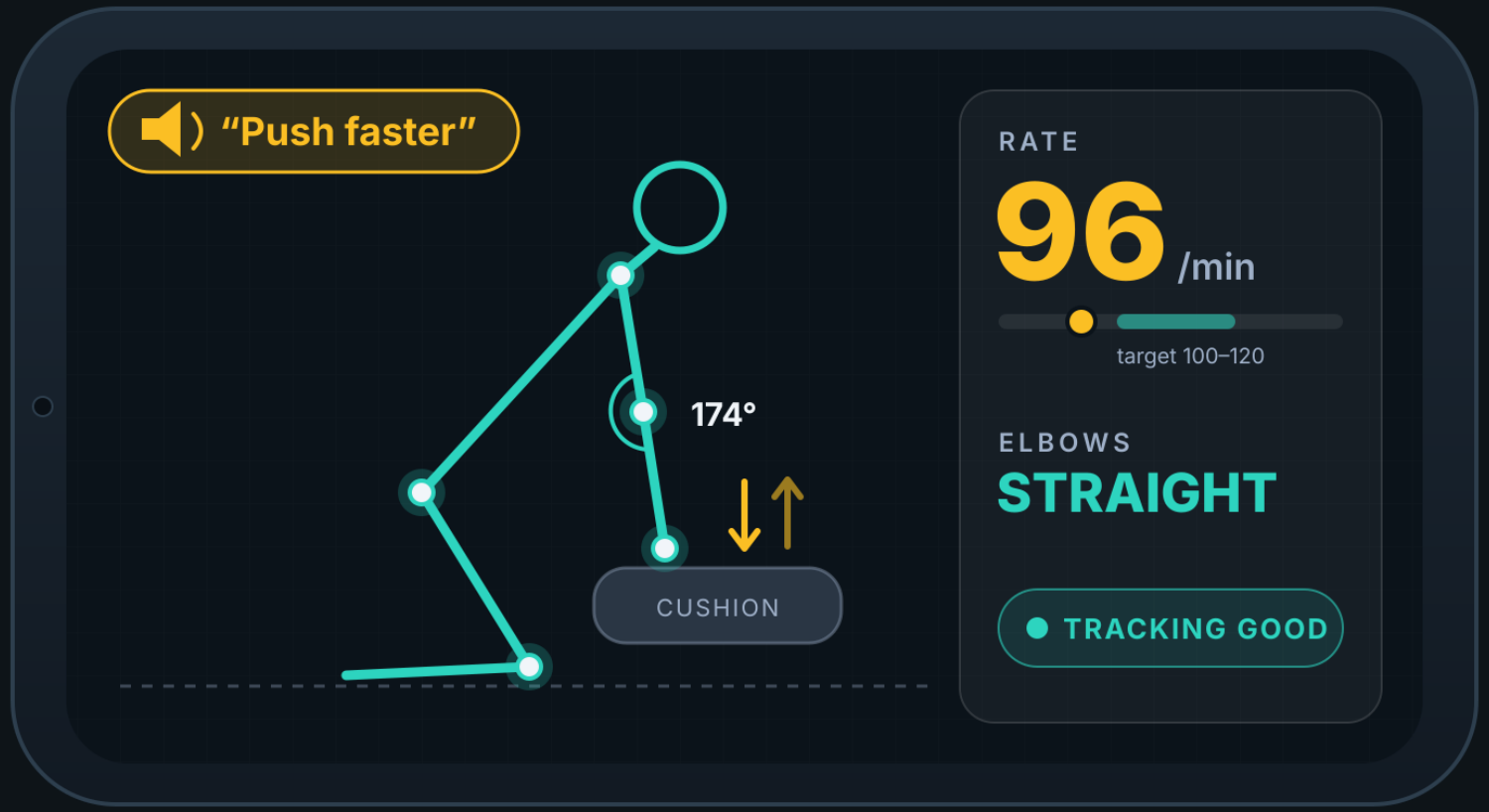
Lybrate survey, 2016 · AHA 2025 Guidelines, Part 12: Resuscitation Education Science

Feedback makes practice stick, but today it needs a sensor manikin

AHA 2025 recommends feedback devices in CPR training, including for lay rescuers (Class 1).

	Sees your technique	Corrects you live	What you need
Video tutorial	No	No	A screen
Metronome app	No	Beat only	A phone
Feedback manikin	Yes	Yes	A sensor manikin
CPR Coach	**Rate, elbows, pauses**	**Spoken, 10 languages**	**A phone**

It watches your arms and says what to fix



01 **Watch**

the camera sees shoulder, elbow and wrist

02 **Count**

rate, elbow bend and pauses

03 **Speak**

one correction, in your language

Illustration · values are examples, not measurements

Camera to voice, every step runs on the phone



phone held side-on

MediaPipe finds the joints

rate, elbow angle, pauses

one correction at a time

recorded clips and a 110/min beat

No internet permission. Nothing is uploaded.

`Kotlin · Jetpack Compose · CameraX · MediaPipe Pose · sherpa-onnx (Whisper, IndicConformer) · Silero VAD`

Practise teaches you; Emergency starts the beat in one tap

PRACTISE

Learn with feedback

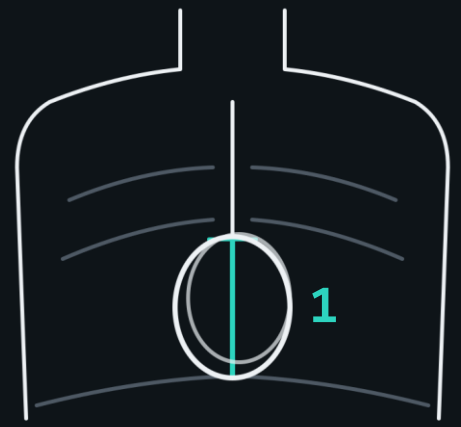
Ran on a phone: 7 real recordings

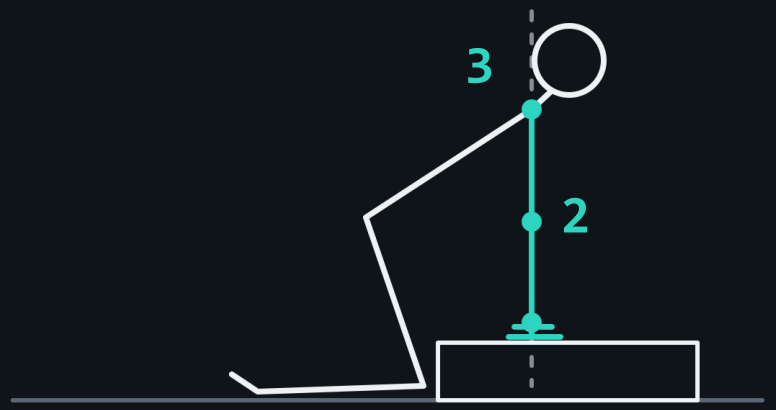
EMERGENCY

Act in seconds

Ran on a phone: steps, beat, dialer

Good CPR needs three things; the app measures one of them





Illustration

1 **Hands on the lower half of the breastbone**

One camera can't see this

2 **Arms straight**

The app measures this

3 **Shoulders over hands**

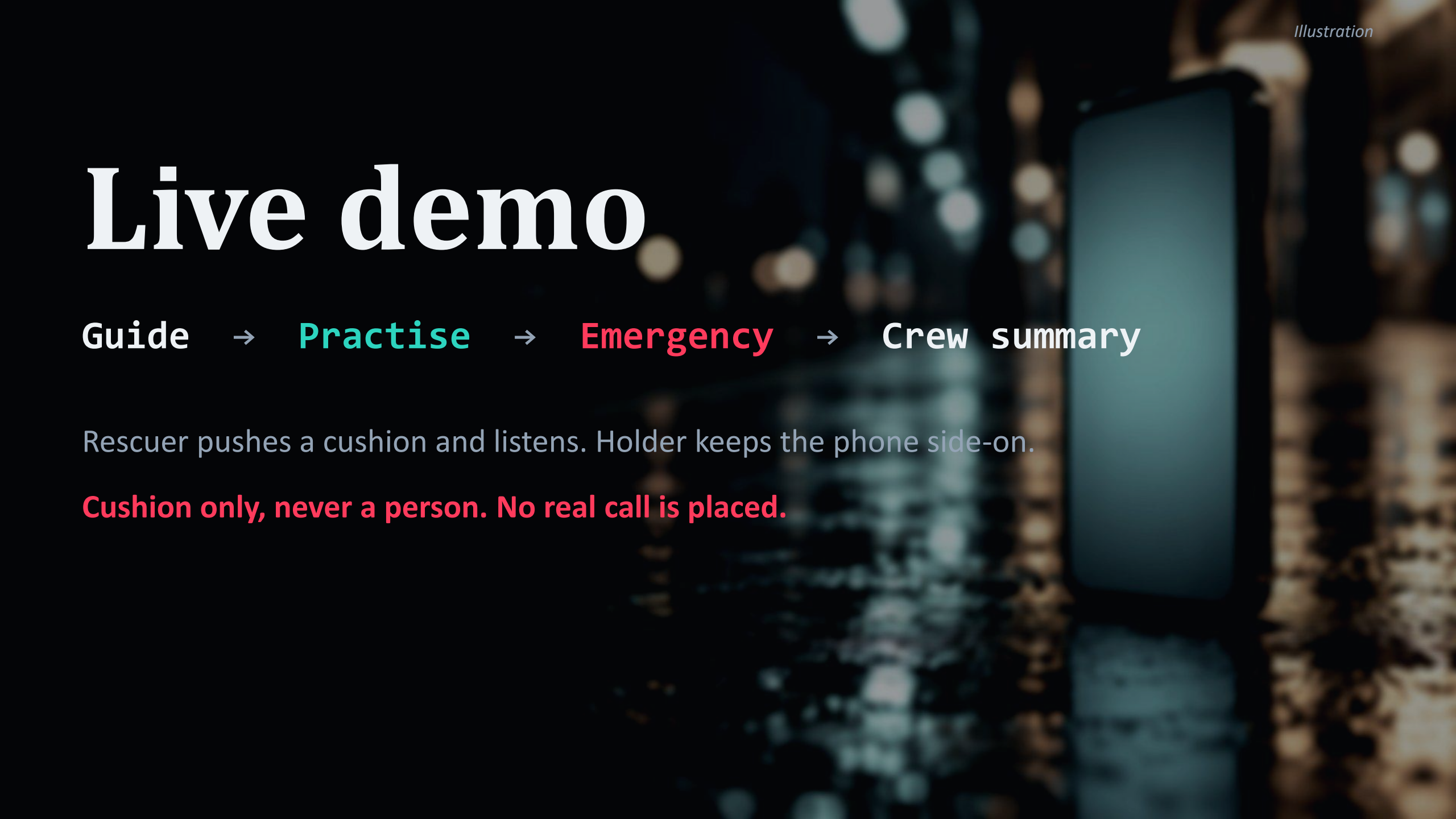
Not measured by the app

It coaches in ten Indian languages, each in its own script

English हिन्दी ಕನ್ನಡ मराठी ગુજરાતી

বাংলা ଓଡ଼ିଆ தமிழ் తెలుగు മലയാളം

English is checked. The other nine are machine-translated and need native-speaker review.



Live demo

Guide → Practise → Emergency → Crew summary

Rescuer pushes a cushion and listens. Holder keeps the phone side-on.

Cushion only, never a person. No real call is placed.

It shows a dash, never a guess, for anything it can't see

IT MEASURES

Push rate

Elbow angle

Pauses

Tracking quality

ONE CAMERA CAN'T

Compression depth

Force and full recoil

Exact hand placement

Clinical effectiveness



Hand-counted accuracy runs come next. Until then, we quote no accuracy figures.

399 tests and 7 real recordings gate every change

399

unit tests passing, 0 failing

7

real phone recordings must replay byte for byte

2

Android phones for on-device testing

14

AI agents on 4 PCs, one lead reviewing every diff

Built in 24 hours by four people and an AI agent swarm. No change merges without passing both gates.

Illustration

Next: count it by hand, then test it with CPR trainers

Then pilot it with NSS units, first-aid clubs and schools.



Not a medical device. In an emergency, call 112 (108 for an ambulance) and start CPR.

Scan for the code and test records

`github.com/VivekTiwary-25/Zero_flux`

Team Zero Flux Vivek · Pratham · Prabal · Moksh